Driving pulley diameter (D_1) = 200mm

Driven pulley diameter (D_2) = ?

Driving pulley speed N_1 = 3000 RPM

Driven pulley speed N_2 = 1500 RPM

$$D_1N_1 = D_2 N_2$$

$$D_2 = \frac{D_1 N_1}{N_2}$$

$$= \frac{200 \times 3000}{1500}$$

$$= 200 \times 2$$

D_2= 400 mm

The diameter of the pulley is 400 mm and the belt should be connected in cross belt method.

BELT SLIP

When power is transmitted through belt driver, the follower of the driver will not rotate at the estimated speed. It will rotate at a lower speed only. The main reason for this defect is slackness of the belt.

Belt slip is the difference between the distance covered by a point on the pulley and the distance covered by a point on the belt per minute. Belt slip is always expressed in percentage.

$$\text{Belt slip} = \frac{\text{Estimated speed} - \text{Actual speed} \times 100}{\text{Estimated speed}}$$

If D_1 and D_2 are the diameters of the pulleys and N_1 and N_2 are their speed in rpm and 'S' is the amount of belt slip in percentage.

$$\text{Velocity ratio} = \frac{N_2}{N_1} = \frac{D_1}{D_2} \times \frac{(100 - s)}{100}$$

EXAMPLE 4

A driving pulley of diameter 120 mm rotates at a speed of 400 rpm. The driven pulley of diameter 80 mm connected by a belt drive rotates at speed of 588rpm. Find the percentage of belt slip.

D_1 = 120 mm D_2 = 80 mm

N_1 = 400 rpm N_2 =?

Driving pulley diameter (D_1) = 120 mm

Driven pulley diameter (D_2) = 80 mm

Driving pulley speed N_1= 400 RPM

Driven pulley speed N_2 = ?

$$D_1N_1 = D_2 N_2$$

$$N_2 = \frac{D_1 N_1}{D_2}$$

$$= \frac{120 \times 400}{80}$$

$$N_2 = 600 \text{ RPM}$$

Estimated speed N_2 = 600 RPM

Actual speed N_2 = 588 RPM

The estimated speed of the driven pulley is 600 rpm. But it rotates at 588 RPM.

$$\text{Belt slip\%} = \frac{\text{Estimated speed} - \text{Actual speed} \times 100}{\text{Estimated speed}}$$

$$= \frac{600 - 588}{600} \times 100$$

$$= \frac{12}{600} \times 100$$

$$= 2$$

Belt slip = 2 %

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